MILITARY SPECIFICATION

DETECTOR, AUXILIARY, X-RAY PROBE,

DT-302/PDR-56

1. SCOPE

1.1 This specification covers requirements for an auxiliary detector, X-ray probe, for use with the portable battery operated, alpha survey instrument, radiac set, AN/PDR-56(). The detector shall use the scintillation detection principle with photo-multiplier tube and shall be capable of detecting soft X-rays (14-21 kiloelectron volts (Kev)) associated with the radioactive decay of plutonium 239 (Pu-239).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901	- Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-D-1000	- Drawings, Engineering and Associated Lists.
MIL-Q-9858	- Quality Program Requirements.
MIL-E-15090	- Enamel, Equipment, Light-Gray (Formula No. 11).
MIL-E-16400	- Electronic Equipment, Naval Ship and Shore: General Specification.
MIL-E-17555	- Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-M-19590	- Marking of Commodities and Containers to Indicate Radioactive Material.
MIL-R-22732	- Reliability Requirements for Shipboard and Ground Electronic Equipment.
MIL-M-23313	- Maintainability Requirements for Shipboard and Shore Electronic Equipment and Systems.
MIL-P-24052	- Preliminary Plans and Test Procedure Requirements for Radiac Instruments.
MIL-M-24100	- Manuals, Orders and Other Technical Instructions for Equipment and Systems.

STANDARDS

MILITARY

MIL-STD-108	- Definitions of and Basic Requirements for Electric and Electronic Equipment.
MIL-STD-167	- Mechanical Vibrations of Shipboard Equipment.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-701	- Preferred and Guidance Lists of Semiconductor Devices.

PUBLICATIONS

NAVSHIPS

NAVSHIPS 93820	- Handbook for the Prediction of Shipboard and Shore Electronic Equipment Reliability.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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3. REQUIREMENTS

3.1 General requirements. - The X-ray probe shall meet the requirements of MIL-E-16400 and the requirements herein unless otherwise authorized by the procuring activity. The design objectives specified herein, including simplicity, reliability, maintainability and quality of the equipment consistent with the needs of the Naval services, shall be considered during all phases of the equipment design, evaluation and production.

3.1.1 The design, construction and assembly of the X-ray probe shall be as specified herein and as shown in the referenced drawings (see 6.2). Where the requirements of this specification and MIL-E-16400 conflict, this specification, including the drawings listed herein, shall govern. Any contractor recommended changes to the design or construction or any proposed modifications of the design or construction, shall be completely described in detail, including drawings of the proposed changes, and submitted to the procuring activity for review and approval prior to the submittal of the preproduction models by the contractor for Government evaluation and approval.

3.1.1.1 Preliminary plans and test procedures. -

3.1.1.1.1 The contractor's preliminary plans and test procedures shall be submitted to the Naval Electronic Systems Command (NAVELECSYSCOM) or procuring activity for review and approval prior to the submittal of the preproduction models for Government test and evaluation. Preliminary plans and test procedures shall be in accordance with MIL-P-24052.

3.1.1.1.2 Copies of preliminary plans, test procedures and the contractor's in-plant test results on his preproduction model tests shall be submitted with the preproduction models when these units are submitted for Government test and evaluation.

3.1.2 Parts and materials. - Except as otherwise specified herein or as authorized by the procuring activity, all parts and materials used shall be in accordance with the requirements of MIL-E-16400. No parts that require special selection from commercial production runs in order to meet the equipment operational requirements shall be used. Cast iron and semisteel shall not be used as materials of construction. Other materials shall be used judiciously with due regard to stress involved and relative importance of the parts in question.

3.1.2.1 Nonstandard parts. - Approval of the use of nonstandard parts shall be obtained as specified in MIL-E-16400. Approval of the use of nonstandard parts must be obtained prior to the submittal by the contractor of preproduction models of the equipment for Government test and evaluation.

3.1.2.2 Preferred semiconductors only in accordance with MIL-STD-701 shall be used in the design and construction of the X-ray probe.

3.1.3 Maintenance and serviceability. - The mechanical design, layout and construction of the X-ray probe shall permit ready assembly and disassembly of the probe for repair and servicing. The layout and assembly of all parts and the location of components shall facilitate calibration, testing and maintenance of the probe by service personnel without requiring special tools or test equipment. The maintainability requirements shall be in accordance with MIL-M-23313.

3.1.3.1 Test points and test features including calibration controls shall be as shown on the drawing (see 6.2) unless otherwise specified by the procuring activity.

3.1.4 Finish. - External surfaces of the probe shall be free of cracks, crevices, indentations or other surface irregularities in which radioactive materials can collect. Outside surfaces shall be prepared and painted in accordance with MIL-E-16400. Enamel used shall be type III, class 1, of MIL-E-15090.

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3.1.4.1 Decontamination. - Surfaces and parts subject to radioactive contamination shall be readily capable of decontamination without recourse to laboratory methods or techniques. Probe surfaces shall be smooth and present a minimum of crevices. Indentification markings engraved on the probe housing or identification plates shall be filled with a nonporous paint of contrasting color to produce a plainly marked, smooth surface.

3.1.5 Tropicalization of the probe shall not be required.

3.1.6 Identification plate and markings. - Identification plate and markings of the probe shall be in accordance with MIL-E.16400. Space shall be included on each probe identification plate for adding the serial number of the individual AN/PDR-56() radiac set modified for use of the specific X-ray probe.

3.1.6.1 Marking of carrying case for radioactive material. - Each carrying case to be supplied with the X-ray probes (see 3.3.1(c)) shall be appropriately marked as required by MIL-M-19590 to indicate the presence of radioactive material, which is part of the AN/PDR-56() radiac set.

3.1.7 Engineering drawings. - Engineering drawings to be supplied by the contractor shall be in accordance with MIL-D-1000.

3.2 Reliability. - The contractor shall conduct the reliability assurance program as required by MIL-E-16400 to assure the X-ray probes reliability requirements specified herein.

3.2.1 In addition, the contractor shall conduct the reliability program specified in MIL-R-22732, including testing as specified in section 4 herein.

3.2.1.1 Reliability level. - Reliability requirements for the X-ray probe shall be 2500 hours mean-time-between-failure (MTBF). Method D of NAVSHIPS 93820 shall be used for prediction of this required MTBF.

3.2.1.2 Quantitative reliability requirements of 1000 hours MTBF shall be demonstrated as specified in 4.4.4.

3.3 General detail requirements.

3.3.1 Equipment composition. - The auxiliary detector X-ray probe shall consist of the following items for each complete unit:

(a) One X-ray probe detector, DT-302/PDR-56.
(b) One common connector, type MC11E-10-6SN, for modifying the existing AN/PDR-56 radiacmeter.
(c) One carrying case with dimensions sufficiently large and partitioned to house the AN/PDR-56 radiac set, including the main probe, auxiliary probe, X-ray probe, shoulder harness, headphones, probe extension handle; two technical manuals for the AN/PDR-56 and two technical manuals for the X-ray probe. Padding used inside the case shall be sufficient to prevent damage to the various units in transit and during shock tests of 4.4.4(g), (h) and (i)).

3.3.2 Environmental conditions. - The X-ray probe shall be capable of normal operation under environmental conditions required by MIL-E-16400 and specified in 3.3.2.1 through 3.3.2.6.

3.3.2.1 Temperature. - Operating temperatures shall be as specified in MIL-E-16400, class 4, except that the low temperature limit shall be minus 20°Centigrade (C).

3.3.2.1.1 Nonoperating temperature requirements shall be as specified in MIL-E-16400.

3.3.2.2 Humidity. - The X-ray probe shall operate satisfactorily and within the performance requirements of 3.5 during and after the humidity tests of MIL-E-16400.

3.3.2.3 Altitude. - The X-ray probe shall operate satisfactorily at pressure ranging from the equivalent of 1-1/2 atmospheres up to the equivalent of 10,000 feet altitude and shall not be damaged by pressures equivalent to 40,000 feet altitude.

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3.3.2.4 Enclosure. -

3.3.2.4.1 Light tightness. - The X-ray probe shall be light tight when tested as required in 4.3.2.1(a).

3.3.2.4.2 Watertightness. - The X-ray probe shall be watertight when tested in accordance with MIL-STD-108 (see 4.3.2.1(b)).

3.3.2.5 Shock. - The X-ray probe and the X-ray probe in the carrying case shall withstand the shock tests specified in 4.4.4(g) and (h).

3.3.2.6 Vibration. - The X-ray probe shall operate satisfactorily when tested as specified in 4.4.4(i).

3.4 Design requirements. - The X-ray probe shall consist of a scintillation crystal assembly, photomultiplier tube detector, and a transistorized amplifier with pulse height (window) selection circuitry (see 6.2).

3.4.1 Scintillation crystal. - A thallium activated sodium iodide tellurium, NaI(Tl), scintillation crystal, type HE, Harshaw Chemical Company, or equivalent, shall be used (1-1/4 inch diameter and 1 millimeter thick). The crystal shall be machined, polished and sealed in an assembly of a round aluminum frame and shall include a 1/8 inch thick optical glass window and a .001 inch aluminum (7 milligrams per square centimeter (mg/cm^2)) entrance window (see 6.2).

3.4.1.1 The crystal assembly shall be optically coupled to the photomultiplier tube by the use of Dow Corning silicone compound QC-2-0057, or equivalent, when approved by the procuring activity.

3.4.1.2 The crystal assembly shall be protected by an aluminized Mylar film cover (.003 inch thick and 12 mg/cm^2 weight) (see 3.4.5.3 and 6.2).

3.4.2 Photomultiplier tube. - Unless otherwise authorized by the procuring activity, the RCA type 4440 photomultiplier tube, or equivalent, shall be used in the X-ray probe.

3.4.2.1 Means shall be provided within the X-ray probe assembly for adjustment of the high voltage supplied to the PM tube from the AN/PDR-56 radiacmeter, such as a tap on a series resistor string. This adjustment is to be used as a coarse gain control and shall be "factory set" and changed only as needed on the replacement of the PM tube.

3.4.3 Circuitry. - The transistorized amplifier pulse height (window) selection circuits within the X-ray probe shall amplify and shape the signals from the PM tube and feed the resulting signal pulses to the Schmitt trigger circuit in the AN/PDR-56 radiacmeter (see 6.2). Semiconductor devices used shall be in accordance with MIL-STD-701.

3.4.3.1 Grain control. - A control shall be provided in the X-ray probe to allow adjustment of the amplifier gain to accommodate the anticipated spread in the scintillation crystals and the PM tube sensitivities.

3.4.3.2 Pulse window control. - The probe circuitry shall include lower and upper discriminator circuits. The lower level discriminator shall reject noise signals with amplitude levels less than 14 Kev equivalent. The upper level discriminator shall reject background noise signals with amplitudes greater than 21 kev equivalent. The "window", or difference between the upper and lower discrimination levels, shall be adjustable by a single control which shall be so designed to center the 14 to 21 Kev Pu-239 X-ray peak within the desired window limits.

3.4.3.3 Output pulses. - The output pulses of the X-ray probe circuitry shall have the required characteristics to drive the Schmitt circuit, or equal, of the AN/PDR-56 radiacmeter without readjustment of any control settings of the radiacmeter circuits.

3.4.4 Electrical requirements. - Nominal operating voltages for the X-ray probe shall be supplied from the high voltage power supply and batteries of the AN/PDR-56 radiacmeter. Voltage tolerances shall be within the tolerances of the voltages supplied by the power supply of the AN/PDR-56 radiacmeter.

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3.4.4.1 Electrical interconnections between the X-ray probe and the radiacmeter shall be the existing retractile coil cord of the radiacmeter which shall be terminated by a Cannon Electric Co. connector, type MC 11E-10-6SN, or equal. A mating hermetic type receptacle (Cannon Electric Co. type MC 14H-10-6PN, or equivalent) shall be provided on the X-ray probe. The pin connections shall be as follows:

Pin number	Function
A	High voltage
B	Unused
C	Signal pulse
D	Instrument ground
E	Plus 3 volt battery supply
F	Unused

3.4.4.2 A protective cap (Cannon Electric Co. type MC-10-60 or equivalent) shall be provided for the connector receptacle. The protective cap shall be affixed adjacent to the connector receptacle on the probe.

3.4.5 Mechanical design (see 6.2). -

3.4.5.1 Form. - The X-ray probe housing shall be cylindrical, with the entrance window and crystal assembly mounted in one end and the connector receptacle and access port on the opposite or handle end of the cylinder (see 6.2.)

3.4.5.1.1 Overall dimensions of the X-ray probe housing shall not exceed 2-3/4 inches diameter by 12 inches long, and the weight shall not exceed 2-1/2 pounds.

3.4.5.2 Access port. - A slotted screw type access port shall be provided in the handle end of the probe housing to permit adjusting of the amplifier gain (Cal. G) and the pulse window width (Cal. W) controls without dismantling the probe.

3.4.5.3 Protective Mylar film cover. - An aluminized Mylar film, .003 inch thick and 1-5/8 inches diameter, density approximately 12 mg/cm^2, shall be installed in the entrance window for protection of the scintillation crystal assembly. The Mylar film shall be aluminized on both sides to prevent light leaks through the film to the PM tube enclosure.

3.4.5.4 Plastic cover. - A clamp-on plastic cover to fit over and protect the window end of the X-ray probe when not in use shall be supplied with each X-ray probe.

3.4.5.5 Probe handle. - The X-ray probe shall be fabricated to permit the use of the existing AN/PDR-56 handle with or without using the handle extension rod.

3.5 Performance. -

3.5.1 Energy range. - The X-ray probe shall selectively measure X-rays in the energy ranges of 14 to 21 Kev.

3.5.2 Sensitivity ranges. - Sensitivity ranges of the X-ray probe/AN/PDR-56 radiacmeter shall be from 0.1 to 200 milligrams per square meter (mg/m^2), equivalent to Pu-239. Ranges of indication shown on the AN/PDR-56 radiacmeter shall be as follows:

Pu-239 concentration (mg/m^2)	AN/PDR-56 scale ranges (cpm)
0.1 - 0.2	0 - 1K
0 - 2.0	0 - 10K
0 - 20	0 - 100K
0 - 200	0 - 1M

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3.5.3 Accuracy.-

3.5.3.1 Conditions.- The following conditions shall apply with respect to accuracy requirements of the X-ray probe:

(a) Thin, uniformly distributed Pu-239 sources with size and alpha energy characteristics equivalent to those of the AN/UDM-7A alpha calibrator shall be used.
(b) The source-to-X-ray probe geometry shall be the same as that used for calibration (see 3.5.4) of the probe.
(c) Background readings shall be subtracted from the actual readings taken of the Pu-239 source.

3.5.3.2 Response.- Response of the X-ray probe shall be linear. Radiacmeter reading at any point over the upper 50 percent of the 0-1K counts per minute (cpm) scale and the upper 90 percent of all other ranges shall be within plus or minus 30 percent of the readings required by the actual Pu-239 concentrations based on linear interpolations of the full scale relationships of 3.5.2.

3.5.3.3 Background response.- The X-ray probe response to normal background radiation, PM tube noise, bright sunlight and probe circuit noise shall be less than 250 cpm indicated on the 1K scale of the AN/PDR-56 radiacmeter.

3.5.3.4 Gamma response.- Meter indications on the AN/PDR-56 radiacmeter with the X-ray probe attached, shall be less than 40K cpm when the probe is placed in a 10 milliroentgens per hour (mr/hr) cobalt-60 gamma field.

3.5.4 Calibration.- The X-ray probe shall be calibrated using the AN/UDM-7A alpha calibration sources (Pu-239). Before calibrating the X-ray probe, the AN/PDR-56 radiacmeter with the alpha probe attached shall be calibrated following the procedures described in the technical manual for the AN/PDR-56. The X-ray probe shall then be attached to the AN/PDR-56, and calibration of the X-ray probe shall be accomplished without readjusting the AN/PDR-56 calibration controls or the high voltage setting of the radiacmeter.

3.5.5 Reproducibility.- Any meter reading on that portion of the scale range on which accuracy requirements apply (see 3.5.3) shall be reproducible to within 15 percent when the probe is placed in an identical field under identical operational and environmental conditions.

3.5.6 Battery life.- The AN/PDR-56 radiacmeter with the X-ray probe attached shall operate as specified for at least 40 hours per set of batteries, including 4 hours intermittent operation of the meter light. Changing of the batteries in the radiacmeter shall not require recalibration of the X-ray probe.

3.6 Technical manuals.- In addition to the requirements for technical manuals in accordance with MIL-M-24100 or revisions to existing technical manuals for the AN/PDR-56 radiacmeter, the information specified in 3.6.1 shall be included in the data for the X-ray probe.

3.6.1 The contractor shall determine and list all the component parts, electronic and otherwise, that can be replaced in the X-ray probe without affecting the calibration of the X-ray probe or the AN/PDR-56 radiacmeter. All such parts shall be listed in the technical manual under the maintenance section.

3.7 Workmanship.- Workmanship shall be as specified in requirement 9 of MIL-STD-454 and shall include requirements stated herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's quality assurance system.- The contractor shall provide and maintain an effective inspection and quality assurance system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the cognizant government inspector for system approval prior to preproduction inspection. Any changes to the approved quality

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assurance system which might affect the degree of assurance required by this specification or other applicable documents shall be submitted to the cognizant inspector and approved in writing prior to use.

4.1.2 Government verification. - All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written system description are being properly applied, and (b) Government product inspection to measure quality of product to be offered for acceptance. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.3 Quality control. - The contractor shall provide and maintain a quality control system acceptable to the Government and approved by the procuring activity for the supplies covered by the contract or order. The system of quality control shall be in accordance with MIL-Q-9858 and this specification.

4.2 Classification of inspection. - The methods of examination and testing of the X-ray probe shall be classified as follows:

(a) Preproduction model inspection.
(b) Quality conformance inspection.

4.3 Preproduction. -

4.3.1 Tests, test equipments and test procedures. - A list of the contractor's proposed tests, test equipments to be used and test procedures shall be furnished by the contractor to the NAVELECSYSCOM or procuring agency, for review and approval 30 days prior to setting up the test equipment for conducting the required preproduction model evaluation tests. Copies of the contractor's tests and test results shall be supplied with the preproduction model when submitted for Government testing and evaluation (see 3.1.1.1).

4.3.2 Preproduction model inspection. - Preproduction model of the X-ray probe shall be examined and tested in accordance with MIL-E-16400, 4.4 herein and as follows to insure conformance of the probes to the performance requirements of this specification, as well as to the design and construction requirements of MIL-E-16400:

(a) Surface examination.
(b) Operating test.
(c) Adequacy of voltages supplied from the AN/PDR-56 radiacmeter.
(d) Heat test.
(e) Enclosure tests.
    (1) Watertightness.
    (2) Light tightness.
    (3) Pressure.
(f) Temperature.
(g) Humidity.
(h) Radio interference and undesired radiation.
(i) Shock and vibration (see 4.4.4 (g), (h) and (i)).

4.3.2.1 Additional tests required. - In addition to the preproduction model tests required by MIL-E-16400 and this specification, the following tests shall be performed:

(a) X-ray probe light tightness test.
    (1) Each probe shall be tested for light tightness, using a light source of a photoflood light bulb (size no. 1) in a suitable reflector placed 2 inches from the X-ray probe window, as well as all other probe surfaces.
    (2) In no case shall the meter deflection of the AN/PDR-56 exceed the combined noise level specified (that is, PM tube plus background plus circuit noise) of less than 250 cpm. Excessive spot patching of the probe aluminized Mylar film window shall be unacceptable as a method of reduction of the required noise level of 250 cpm.
(b) X-ray probe water tightness tests shall be in conformance to the requirements of MIL-STD-108.
(c) Battery life tests as required to verify that the battery life of the AN/PDR-56 radiacmeter with the X-ray probe attached shall meet the minimum operational requirements of 40 hours, including 4 hours of intermittent use of the meter light.

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(d) Such tests as are required to insure conformance of the X-ray probe to the requirements of operating at altitudes specified in 3.3.2.3 herein.
(e) Gamma response - Tests using a test source of 10 mr/hr cobalt-60 gamma field shall be conducted. Meter readings on the AN/PDR-56 radiacmeter with the X-ray probe shall be less than 40K cpm with the X-ray probe placed in this gamma field.

4.3.3 Reliability analysis and prediction shall be conducted in accordance with MIL-R-22732 (see 3.2.1.1).

4.3.4 Preproduction model testing and evaluation will be performed at a Government agency designated by the procuring activity. Prior to submission of the preproduction model by the contractor for Government evaluation, the contractor shall submit to the procuring activity concerned copies of data of tests conducted and results obtained by the contractor to insure that the X-ray probes do in fact meet the design and performance requirements of this specification and MIL-E-16400.

4.4 Quality conformance inspection. -

4.4.1 General inspection. - The general examination and tests shall consist of the inspection of MIL-E-16400 and the following:

(a) Reliability testing.
(b) Production inspection.
(c) Production control inspection.
(d) Environmental tests.

4.4.2 Reliability testing. - Reliability testing shall be performed in accordance with test level 2 of MIL-R-22732 with the following modifications thereto:

(a) Temperature cycling, minus 20°C to plus 50°C.
(b) Temperature cycling of the X-ray probe to the extent required to insure that the X-ray probe operates as required over the temperature range specified in 4.4.2(a).
(c) Vibration, 33 cycles per second (cps) single amplitude, .010 inch, type I, of MIL-STD-167.
(d) Input voltages as supplied from the AN/PDR-56 radiacmeter.

4.4.2.1 Performance parameters to be monitored during the test cycle of 4.4.2 shall include the following:

(a) Calibration of the X-ray probe (see 3.5.4).
(b) Accuracy of the X-ray probe on all four ranges (see 3.5.3.2).
(c) Noise and background response of the X-ray probe (see 3.5.3.3).
(d) Gamma response of the X-ray probe (see 3.5.3.4).
(e) Battery life of the AN/PDR-56 radiacmeter when using the X-ray probe.

4.4.3 Production tests. - Production tests shall be as specified in MIL-E-16400 and the following tests:

(a) Each X-ray probe shall be tested in accordance with 4.3.2.1 (a) for conformance to the light tightness requirements for the probes.
(b) Each X-ray probe shall be tested for calibration at one point between 30 and 80 percent of full scale on each of the four ranges of the AN/PDR-56.

4.4.4 Production control tests. - Production control tests shall include the tests of MIL-E-16400 and the following tests:

(a) Check of the noise and background response and limitations (see 3.5.3.3).
(b) Check of the X-ray probe calibration at one point on the 0 - 100K cpm scale between 20 and 90 percent of full scale. This check point to differ from that of the similar production calibration testing requirements by at least 30K (cpm). Use of the AN/UDM-7A calibrator recommended.

(c) A check of the X-ray probe gamma radiation response requirements of 3.5.3.4.
(d) Battery life check requirements of 3.5.6.

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(e) X-ray probe enclosure requirements of 3.3.2.4.
(f) Reliability of 1000 hours MTBF(see 3.2.1.2).
(g) Shock.- The X-ray probe shall withstand the type A shock tests of MIL-S-901, except that the hammer drops shall be 1, 2 and 3 feet in lieu of 1, 3 and 5 feet specified. The probe shall be energized during the shock tests and after each blow shall continue to operate satisfactorily without circuit adjustments or replacement of parts. There shall be no evidence of collision between parts as a result of these tests. Changes in the probe calibration as a result of the shock tests shall not exceed the accuracy requirements of this specification (see 3.5.3).
(h) The X-ray probe in the carrying case shall withstand type A shock tests of MIL-S-901. There shall be no evidence of collision between the X-ray probe and other components of the radiac set in the case. The X-ray probe shall operate satisfactorily after the shock tests without recalibration or replacement of parts. The 5 foot hammer drop only need be made for this test.
(i) Vibration.- The X-ray probe shall operate satisfactorily during and after the type I vibration tests of MIL-STD-167. Accuracy requirements of the probe (see 3.5.3) shall be maintained during and after the test runs. There shall be no evidence of collision between component parts of the probe resulting from the vibration tests.

4.4.4.1 Sampling for production control tests.- Samples for production control tests shall be selected at random from equipments that have passed production tests. Sampling size shall be determined by the use of table I.

Table I - Sampling for production control tests.

Number of units in lot	Number of units in sample	Number of units nonconforming on any tests	
		Acceptance number	Rejection number
15 and under	3	0	1
16 to 40	5	0	1
41 to 110	7	0	2
111 to 300	10	1	2
301 and over	15	2	3

4.4.4.2 Each of the sample X-ray probes selected in accordance with 4.4.4.1 shall be subjected to all the tests specified in 4.4.4 to determine compliance with this specification. Failure to conform shall be counted as a defect, and the X-ray probe shall be rejected. If the number of nonconforming X-ray probes in the sample equals or exceeds the rejection number for the sample, the entire lot represented by the sample shall be rejected.

4.4.5 Environmental tests.- Environmental tests shall include temperature, humidity, water tightness (enclosure) and electromagnetic (radio interference) tests of this specification and MIL-E-16400.

4.4.5.1 Sampling for environmental tests.- The units used for environmental tests shall be units drawn from the units passing production control tests on the basis of one unit per 50 production control samples, except that not more than one unit per 100 production units shall be drawn for environmental testing.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation.-

5.1.1 X-ray detector.-

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5.1.1.1 Preservation and packaging.- Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.2 Domestic shipment and storage or overseas shipment.- The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The equipment, accessories, technical publication and repair parts shall be preserved and packaged level A or C, packed level A, B or C and marked as specified in MIL-E-17555.)

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Preservation, packaging, packing and marking requirements if other than 5.1 (see 5.2).

6.2 Reference and guidance drawing.- U.S. Naval Radiological Defense Laboratory Drawing, NU-63-606, "PU-239 X-Ray Probe for AN/PDR-56", is for information and guidance purposes only.

Preparing activity:
Navy - SH
(Project 6665-N227SH)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD, FORM 1 APR 63 1426

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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